

RectChr GUI Manual

GUI 1.50C · Engine RectChr 1.50 · hewm2008

A cross-platform desktop interface for RectChr (a circos-like, multi-level chromosome visualization tool). The GUI writes the configuration; the Perl engine renders.

2. Interface (three-column workbench)

Toolbar and the File/View/Help menu buttons; every function has a menu entry.

Area	Content
Left	Data files File1..FileN (.gz, dual-entry sync, 20-row preview)
Center	Preview canvas (white paper) + refresh/auto-refresh/export + stale hint
Right	Parameters: Global / trackALL / Track N / Parameters overview, Track manager card on top
Bottom	Run log (engine output and errors)

Menus: File = New / Open / Save / Export SVG-PNG-PDF; View = show-hide the three panels + Reset layout; Help = manuals (zh/en) + About.

3. Quick start (five steps)

- Add data: left panel Add file; format Chr Start End Value1 ..., .gz supported
- Minimal config: File1 alone is enough (default plot_type = heatmap)
- Run: press Run; or enable Auto refresh and just edit parameters
- Tune: edit in the right-side tabs; a stale hint shows while re-rendering
- Export: Export SVG / Export PDF on the preview strip, or Export PNG in the File menu

Minimal heatmap example

```
SetParaFor=global
File1 = ./data.gz
— one data file, everything else defaults.
```

4. Data format & paths

Tab/space separated text (.gz supported): Chr Start End Value1 The first three columns locate the region; each extra column feeds one track. NA skips a region. show_columns picks columns (e.g. File1:4,7); by default layer N reads File1 column N+3. Relative paths resolve against the conf directory; legacy parameter names are mapped automatically.

5. Parameter reference (by category)

Generated from the GUI schema; defaults and ranges are the engine's real values.

Parameter	Description	Default	Range	Notes
canvas_height_ratio	Canvas height ratio	—		Overall canvas height ratio
canvas_width_ratio	Canvas width ratio	—		Overall canvas width ratio
canvas_angle	Canvas angle	0		Rotation angle of the exported PNG (degrees)
canvas_body	Canvas body	1200		Main canvas size (default 1200)
canvas_margin_bottom	Canvas margin bottom	30		Bottom margin of the canvas (default 30)

canvas_margin_left	Canvas margin left	100		Left margin of the canvas (default 100)
canvas_margin_right	Canvas margin right	80		Right margin of the canvas (default 80)
canvas_margin_top	Canvas margin top	55		Top margin of the canvas (default 55)
Parameter	Description	Default	Range	Notes
chr_scale_ratio	Chr scale ratio	—		Scale chromosome lengths by this ratio
chr_order	Chr order	—		Chromosome order / filter list, e.g. chr1,chr2
chr_zoom_region	Zoom region	—		Zoom into a region, format chr2:1000:5000
chr_orientation	Chr orientation	vertical		vertical or horizontal chromosome layout
chr_spacing_ratio	Chr spacing ratio	0.2		Spacing ratio between chromosomes
chr_label_rotation	Chr label rotation	0		Rotation angle of chromosome labels
chr_label_size_ratio	Chr label size ratio	0.3		Font size ratio of chromosome labels
chr_label_shift_x	Chr label shift X	0		Horizontal shift of chromosome labels
chr_label_shift_y	Chr label shift Y	0		Vertical shift of chromosome labels
chr_order_reverse	Reverse chromosome order	—		Reverse the chromosome display order; setting it takes effect, empty = off (note: the engine currently parses the value as a comma-separated chromosome list, so writing 1 only affects a chromosome literally named 1)

Parameter	Description	Default	Range	Notes
title	Title text	—		Main figure title
title_color	Title color	—		Color of the main title
title_size	Title size ratio	—		Font size ratio of the main title
title_shift_x	Title shift X	—		Horizontal offset of the title
title_shift_y	Title shift Y	—		Vertical offset of the title

Parameter	Description	Default	Range	Notes
axis_tick_unit	Axis tick unit	—		Unit of the coordinate axis ticks (bp/Kb/Mb/...)
axis_text_angle	Axis text angle	—		Rotation angle of the axis tick text
xaxis_tick_show	Show X-axis ticks	1		Show (1) or hide (0) X axis tick labels
axis_tick_num	Axis tick number	10		Number of axis ticks (max 10 parts by default)

axis_tick_interval	Axis tick interval	—		Interval between axis ticks
xaxis_shift_y	X axis shift Y	0		Vertical shift of the X axis
axis_tick_precision	Legend value precision	—		Decimal digits of the colorbar/legend value labels; does NOT affect axis ticks (use yaxis_tick_precision for the Y axis)
yaxis_tick_show	Show Y ticks	0		Show (1) or hide (0) Y axis tick labels
xaxis_tick_direction	X tick direction	—		X axis tick direction: Down (default) or Up
yaxis_tick_num	Y tick number	5	≥ 2	Number of Y axis ticks per track; default 5 (2 when the track is <20px tall, auto from height below 90px)
yaxis_tick_precision	Y tick precision	—	$\geq 0 \leq 10$	Decimal digits of Y axis tick labels; auto when unset: 0 if the Y range is integral, 2 if YMax has >4 decimals, else 1
yaxis_tick_unit	Y tick interval	—		Numeric interval between Y axis ticks; auto-divided from the Y range
yaxis_tick_strokewidth	Y tick stroke width	1	≥ 0	Stroke width of the Y axis ticks, default 1
yaxis_tick_direction	Y tick direction	—		Flip the Y tick direction; the engine acts on the key merely existing (its Down/Up value is ignored), empty = default
xaxis_tick_color	X tick color	—		Color of the X axis ticks, black by default
Parameter	Description	Default	Range	Notes
padding_ratio	Track padding ratio	0	$\geq 0 \leq 1$	Vertical spacing ratio between adjacent tracks
track_height	Track height	20	≥ 5	Height of this track
track_num	Track count	—		Number of tracks (layers); auto-inferred from File1 columns by default
background_color	Background color	#B8B8B8		Background color of this track
background_show	Show background	—		Draw this track's background: 1 = shown (default), 0 = hidden; the engine suggests 0 for LinkS
bg_end_arc	Bg end arc	1		Draw arcs at chromosome ends of the background
bg_end_offset	Extend end curves	—		Extend the chromosome end curves outward; setting it takes effect, empty = off
bg_end_arc_division	Bg end arc division	—		Division of the end curve arc
track_bg_height	Track bg height	1	≥ 0	Height of this track's

_ratio	ratio			chromosome background as a proportion of track_height; decimals allowed (e.g. 0.35), 1 when unset
track_shift_x	Track shift X	0		Horizontal shift of the whole track
track_shift_y	Track shift Y	0		Vertical shift of the whole track
track_text_overlap	Text overlap	—		0 = allow overlap; N = at most N overlapping texts
colormap_legend_layout	Legend layout	1		Legend layout mode (0-13)
colormap_legend_gap	Legend gap	4.5		Gap between the legend and the track (engine default 4.5)
colormap_gradient_gap	Gradient gap	—		Gap between adjacent gradient blocks (default 0)
colormap_legend_sizeratio	Legend size ratio	1.0		Size ratio of the gradient legend (0 hides it)
Parameter	Description	Default	Range	Notes
log_p	-log10 value	0		Apply -log10 transformation to values
cap_max_value	Cap max value	—		Truncate values above this maximum
cap_min_value	Cap min value	—		Truncate values below this minimum
Ymin	Y min	—		Manual minimum of the display range
Ymax	Y max	—		Manual maximum of the display range
show_columns	Show columns	—		Columns to plot, e.g. File1:4 or File2:4,5
as_flag	Treat column as flags	—		Treat this column as text rather than numbers; setting it (any value) takes effect, leaving it empty disables it
upper_outlier_ratio	Upper outlier ratio	0.95		Values above this quantile use the max color
lower_outlier_ratio	Lower outlier ratio	0		Values below this quantile use the min color
hist_No_UD	Histogram: hide U/D ticks	—		Suppress the upper/lower tick marks of a two-column histogram (internal); setting it takes effect, empty = off
Parameter	Description	Default	Range	Notes
label	Track label	—		Label text of this track
label_color	Label color	—		Color of the track label
label_size	Label size	1.0	≥ 0	Font-size scale ratio of the track label (default 1.0)
label_angle	Label angle	—		Rotation angle of the track label

label_shift_x	Label shift X	—		Horizontal shift of the track label
label_shift_y	Label shift Y	—		Vertical shift of the track label
Parameter	Description	Default	Range	Notes
track_geom_shape	Shape type	—	$\geq 0 \leq 20$	Shape type, comma-separated, one per value index: 0-13 basic geometry (circle, square, triangle, ...), 14-19 half-circle and half-ellipse variants, 20 two facing half circles) (centromere style); items outside 0-20 are dropped by the engine, which then uses its automatic shape for that index
link_direction	Link direction	—		Direction of link start/end (UpDown, DownUp, ...)
link_linestyle	Link line style	—		Straight line or bezier curve
track_geom_shape_size	Shape size ratio	1.0		Size ratio of shapes
track_text_size	Text size ratio	1.0		Font size ratio of texts
track_point_size	Point size ratio	1.0		Size ratio of points
track_text_angle	Text angle	0		Rotation angle of texts
track_text_anchor	Text anchor	—		Text anchor (start/middle/end)
line_colors_conf	Line colors file	—		Custom line color configuration file
link_uniform_height	Link uniform height	—		Give all links the same height
cutoff_color	Cutoff line color	—		Color of the cutoff line
cutoff1_color	Cutoff line 1 color	—		Color of the first cutoff line
cutoff2_color	Cutoff line 2 color	—		Color of the second cutoff line
cutoff_y	Cutoff line Y	—		Y position of the cutoff line (docs name: cutoff_value)
cutoff1_y	Cutoff line 1 Y	—		Y position of the first cutoff line
cutoff2_y	Cutoff line 2 Y	—		Y position of the second cutoff line
Parameter	Description	Default	Range	Notes
colormap_low_color	Gradient low color	#006400		Color for the lowest value
colormap_conf	Colormap file	—		Custom value-to-color mapping file, e.g. P1="#FE0808"
colormap_brewer_name	Brewer palette	—		Preset palette name, e.g. GnYlRd (numeric) or Paired (categorical)
colormap_nlevels	Color levels	8	$\geq 3 \leq 255$	Number of color gradient levels

colormap_reverse	Reverse gradient colors	—		Reverse this track's gradient colours; the engine acts on the key merely existing (writing 0 still reverses), empty = off
colormap_mid_color	Gradient mid color	#FFFF00		Color for the middle value
colormap_high_color	Gradient high color	#FF0000		Color for the highest value
colormap_legend_show	Show gradient legend	1		Show the gradient legend: 1 = shown (default), 0 = hidden; also hidden when the legend size ratio is 0
colormap_legend_shift_x	Legend shift X	0		Horizontal shift of the gradient legend
colormap_legend_shift_y	Legend shift Y	0		Vertical shift of the gradient legend
Parameter	Description	Default	Range	Notes
stroke-width	Stroke width	1	≥ 0	SVG stroke-width
fill-opacity	Fill opacity	1	$\geq 0 \leq 1$	SVG fill-opacity
font-family	Font family	Arial		SVG font-family
bg_stroke_color	Bg stroke color	—		Stroke color of the background
stroke-opacity	Stroke opacity	1	$\geq 0 \leq 1$	SVG stroke-opacity
font-size	Font size	—	≥ 0	SVG font-size
fill	Fill color	green		SVG fill color
text-font-size	Text font size	—	≥ 0	SVG text font size
bg_stroke_width	Bg stroke width	0		Stroke width of the background

6. Plot types

13 plot types; switch in the Track tab's plot_type combo — the form follows.

Heatmap (heatmap)

区域统计值映射为颜色格子，基因组密度/Fst/深度等最常用类型

Key params: colormap_low_color、colormap_mid_color、colormap_high_color、colormap_nlevels、Ymax、Ymin

```
plot_type = heatmap
```

Highlights (highlights)

区间背景着色，常用于标记选择区域、QTL 区间、着丝粒等

Key params: background_color、track_bg_height_ratio

```
plot_type = highlights
```

Scatter / point (point)

散点图，GWAS 曼哈顿图的常用画法

Key params: cutoff_y、cutoff_color、log_p、track_point_size

```
plot_type = point
```

Line (line)

折线/曲线，展示连续值走势

Key params: track_height、Ymax、Ymin

```
plot_type = line
```

Histogram (hist)

柱状图；show_columns 给两列时可画上下对称双柱

Key params: show_columns、track_height、yaxis_tick_show

```
plot_type = hist
```

Text (text)

在对应区域显示文字（基因名/标记名等）

Key params: track_text_size、track_text_angle、track_text_anchor

```
plot_type = text
```

Shape (shape)

按区域绘制几何形状（0-13 常规几何形，14-19 半圆/半椭圆组合，20 两个相向的半圆）(着丝粒样式；track_geom_shape 按取值序号用逗号分隔填写)

Key params: track_geom_shape、track_geom_shape_size

```
plot_type = shape
```

Ridgeline (ridgeline)

山脊图，展示 QTL/分布的堆叠形态

Key params: track_height

```
plot_type = ridgeline
```

Pairwise link (PairWiseLink)

两个基因组/区间之间的彩虹连接线（共线性/变异）

Key params: link_direction、link_linestyle、link_uniform_height

```
plot_type = PairWiseLink
```

Self link (LinkS)

染色体自身区域之间的连接

Key params: link_direction、link_linestyle

```
plot_type = LinkS
```

Animated heatmap (heatmapAnimated)

动态 SVG 热图，导出后用浏览器打开可播放动画

Key params: colormap_*, Ymax, Ymin

```
plot_type = heatmapAnimated
```

Animated histogram (histAnimated)

动态 SVG 柱状图，浏览器打开可播放动画

Key params: track_height、show_columns

```
plot_type = histAnimated
```

7. FAQ

- Perl not found (Windows): unzip portable Strawberry Perl into gui_runtime/perl/, or keep Perl on PATH.
- Preview not changing: with Auto refresh ON the canvas re-renders automatically; otherwise press Refresh preview / F5.
- Auto refresh failed: red hint on the canvas; details in the Run log.
- Animations: animated plots show a static frame in the preview — open the exported SVG in a browser.
- Closed a panel: re-enable it in the View menu; Reset layout restores defaults.
- Legacy confs: old parameter names map automatically.

8. Contact

- Email: hewm2008@gmail.com / hewm2008@qq.com
- Home: <https://github.com/hewm2008/RectChr>
- QQ Group: 125293663